SPE-229628-MS

Study on Casing Deformation Mechanism and New Prevention Measure During Hydraulic Fracturing of Shale Reservoir

Qianli Lu, Xinwei Gong, and JianChun Guo, State Key Laboratory of Reservoir Geology and Development Engineering - Southwest Petroleum University; Ziwei Zhang, China University of Petroleum, Beijing

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

"Large-scale, high-intensity, precision-cutting"volume fracturing enables efficient shale reservoir development,but causes CD (CD) during fracturing, forcing abandonment of fracturing sections and increasing costs. In Sichuan Basin, a key "Gas Daqing" area, CD rates exceed 30% in Weiyuan and 60% in Luzhou blocks, severely impacting wellproductivity. Prevention measures like enhanced casing strength and reduced construction scale are unsatisfactory, making new CD prevention method a critical research hotspot in the industry.This study clarified that casing shear deformation caused by fracture slip is the main reason ofCD, and a three-dimensional fracture slipping finite element model integrating casing-cement ring-formation was developed. The influences of natural fracture characteristics, rock mechanical properties, and in-situ stress differences on CD were explored. Based on which, the non-cementing fracturing technology with outer casing packerwas proposed to prevent CD, with feasibility verified by finite element method.Results show that during fracturing, natural fractures are activated. As fracturing fluid is injected, fluid pressure within fractures increases, friction force decreases, causing shear slippage, resulting in local stress concentration and deformation of casing. When the angle between natural fracture and minimum horizontal principal stress is 45°, fracture slip risk and CD magnitude reach maximum.CD worsens with larger differences between horizontal principal stress, larger Young's modulus of rock, and higher fracture fluid pressure. Therefore, theNCF is proposed to reduce the force transmitted from rock to casing and prevent CD.By leaving a sufficient gap for fracture slip between casing and formation, direct contact between rock and casing after fracture slippage is prevented to controlCD. Simulation results show that compared with cementedcasing, this method reduces the maximumcasing load by 81.2%, alleviates local stress concentration, and reduces CDrisk.The Geological monitoring data indicated that CD occurred at well depth of 3090min Well HX-1 of W202 block during fracturing, and the maximumCD magnitude by logging was 31.8mm. The CD behaviors ofcementedcasing and NCF were simulated based on the data of this well. The maximum CD magnitude of the former is 30.58mm, with 3.8% error from logging date, indicating that the finite element model of fracture slip is accurate.The maximum CD magnitude of the latter is 0.22mm, which won't affect subsequent construction, showing CD prevention by NCF is feasible.Research canprovide reference for preventing CD caused by fracture slip.

Key words: shale reservoir, fracture slip, casing deformation, prevention method, non-cementing fracturing with external packer

Introduction

As the development of conventional oil and gas resources gradually faces bottlenecks, unconventional oil and gas resources have become a crucial strategic alternative due to the abundant reserves(Hu and Bao 2013; Li et al. 2022). In the development of deep shale gas, the horizontal well volumetric fracturing technology characterized by "large displacement + dense clustering + temporary plugging and diversion" has emerged as the mainstream technique. However, in complex structural areas with high stress differentials and well-developed natural fractures, this high-intensity fracturing mode often leads to severe CD (CD) issues. Statistics indicate that the CD rates in the Weiyuan and Luzhou blocks of the Sichuan Basin exceed 30% and 60%, respectively, not only causing downhole tool jamming and operational interruptions but also significantly compromising wellbore longevity and development efficiency (Liu et al. 2020; Liu 2021; Wang and Zhang 2022; Zhou 2022). Therefore, a thorough understanding of CD mechanisms and the development of effective prevention and control technologies hold substantial engineering significance.

Existing research has primarily explored CD through numerical simulations and laboratory experiments(Li et al. 2025; Lin et al. 2024; Wang et al. 2025). Yuan (Yuan et al. 2023) utilized a self-developed physical simulation system for fracturing-induced CD to confirm the shear deformation mechanism caused by natural fracture slippage, though the experimental scale limited its field applicability. In terms of numerical simulations, Xu(Xu et al. 2018)established a two-dimensional fracture model and found that CD is prone to occur at the intersection of hydraulic fractures and natural fractures. Fei Yin(Yin et al. 2018) further revealed the bending effect of fracture slippage on casing through a three-dimensional model. Previous studies have investigated influencing factors of CD from multiple perspectives, which can be broadly categorized into two groups.Regarding geological factors, Meng(Meng et al. 2020) developed a two-dimensional plane strain model for fracture shear slippage and observed that greater in-situ stress differentials and fracture dip angles lead to increased slippage and CD. Hu(Hu et al. 2022) further demonstrated that longer fractures exhibit higher slippage probabilities, whereasLiu(Liu et al. 2019) argued that smaller fault angles result in greater slippage, revealing some contradictions in research conclusions. In terms of engineering factors, it is widely acknowledged that higher operational intensity and scale exacerbate CD. Shi(Shi et al. 2023) pointed out that larger gaps in the cement sheath reduce casing-cement contact and increase CD displacement. Zhao(Zhao et al. 2019) and Bois(Bois et al. 2011) suggested that cement layers with higher elastic moduli, due to their excessive stiffness, fail to accommodate deformation synergistically, thereby aggravating CD. To mitigate CD, numerous scholars have proposed prevention methods focusing on enhancing casing collapse resistance and reducing the transfer of fracture slippage forces to the casing. Existing approaches, such as improving casing strength(Shi et al. 2018), optimizing cement slurry properties(Li et al. 2017), adopting novel cement slurry systems(Xi et al. 2023), and adjusting process parameters(Zhang et al. 2023) have yet to fully resolve the issue.

To address this challenge, this study systematically analyzes the influence patterns of geological and engineering factorsby establishing an integrated 3D model of the formation-cement sheath-casing system, innovatively proposes an uncemented fracturing technology using external casing packers and validate the effectiveness of the new technology. The research findings provide critical theoretical support and technical references for the safe and efficient development of shale gas.

Mechanism of Casing Deformation

Shale reservoirs typically exhibit relatively well-developed natural fractures. During hydraulic fracturing, fracturing fluid infiltrates weak planes such as natural fractures, leading to an increase in fluid pressure within the fractures and a reduction in frictional resistance. This process activates the natural fractures, inducing slip (Fig. 1). The forces generated by the slip of natural fractures are transmitted to the casing through the cement sheath. When these forces exceed the casing's collapse resistance strength, deformation occurs.

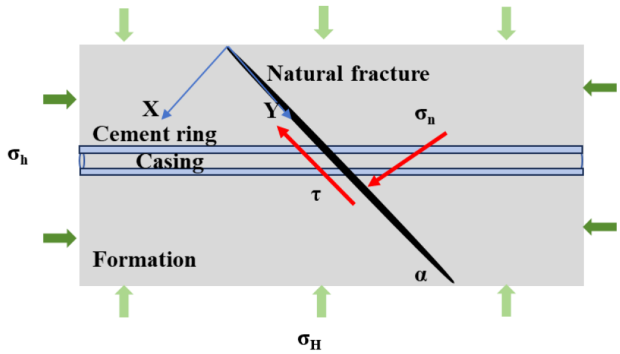


Figure 1—Schematic diagram of CD caused by fracture slipping.

As shown in Fig. 1，the normal stress (σ_n) and shear stress (τ) on the fracture surface before fracturing are:

$$\sigma_n = \frac{\sigma_H + \sigma_h}{2} - \frac{\sigma_H - \sigma_h}{2}\cos 2\alpha \tag{1}$$

$$\tau = \frac{\sigma_H - \sigma_h}{2}\sin 2\alpha \tag{2}$$

During fracturing operations, the fracturing fluid enters and props open natural fractures. At this stage, the fracture surfaces mainly experience rock contact pressure and fracturing fluid pressure in the x-direction, while being primarily subjected to fracture surface friction and wellbore shear forces in the y-direction. That is：

$$\sigma_x = \sigma_n - P_f \tag{3}$$

$$\sigma_y = \tau - f \tag{4}$$

The mechanical condition for natural fracture slip is given by:

$$\tau > C_0 + \mu\sigma_x = C_0 + \mu(\sigma_n - P_f) \tag{5}$$

where:τ is the shear stress on the natural fracture surface, Pa; σ_n is the normal stress on the natural fracture surface, Pa; σ_H is the maximum horizontal principal stress, Pa; $\sigma_h$is the minimum horizontal principal stress, Pa; α is the angle between the natural fracture and the maximum horizontal principal stress,°; C_0 is the cohesive strength of the natural fracture, Pa; μ is the friction coefficient of the natural fracture surface, dimensionless; P_f is the fluid pressure inside the natural fracture, Pa.

Finite Element Model of Casing Deformation

Physical Model Establishment

Based on the physical process of CD, a three-dimensional finite element model integrating formation, cement sheath, and casing for fracture slip analysis was established, as shown inFig. 2. According to Saint-Venant's principle (Guo et al. 2018), boundary effects can be neglected when the formation boundary dimensions exceed 5-6 times the wellbore size. Therefore, the model dimensions were set to 100m×100m×100m, with an initial fracture surface of 30m×30m×1mm. The cement sheath outer diameter was 259.7mm, while the casing outer and inner diameters were 139.7mm and 118.4mm respectively(Fig. 2). Material and geological parameters were configured according to actual working conditions (Table 1).

In this model, formation stresses were applied as predefined fields to simulate the original in-situ stress state. Casing internal pressure and fracture fluid pressure were set based on actual operational parameters to simulate fracturing fluid injection conditions during hydraulic fracturing. This approach enables automatic calculation of fracture slip displacement and CD, eliminating the need to preset fracture slip displacement for CD analysis as required in previous studies.

Table 1—Parameters of the model.

Parameter	Value	Parameter	Value
Young's modulus of formation rock/GPa	26.6	Poisson's ratio of rocks/dimensionless	0.26
Young's modulus of cement ring/GPa	20	Cement ring Poisson's ratio/dimensionless	0.25
Young's modulus of the casing/GPa	200	Casing Poisson's ratio/dimensionless	0.3
Maximum horizontal principal stress/MPa	85	Horizontal minimum principal stress/MPa	79
Fluid pressure in the fracture/MPa	88	Pressure inside the casing/MPa	88
Rocks density /kg/m³	2500	Casing density/kg/m³	7800
Cement ring density/kg/m³	1800	Vertical principal stress/MPa	82

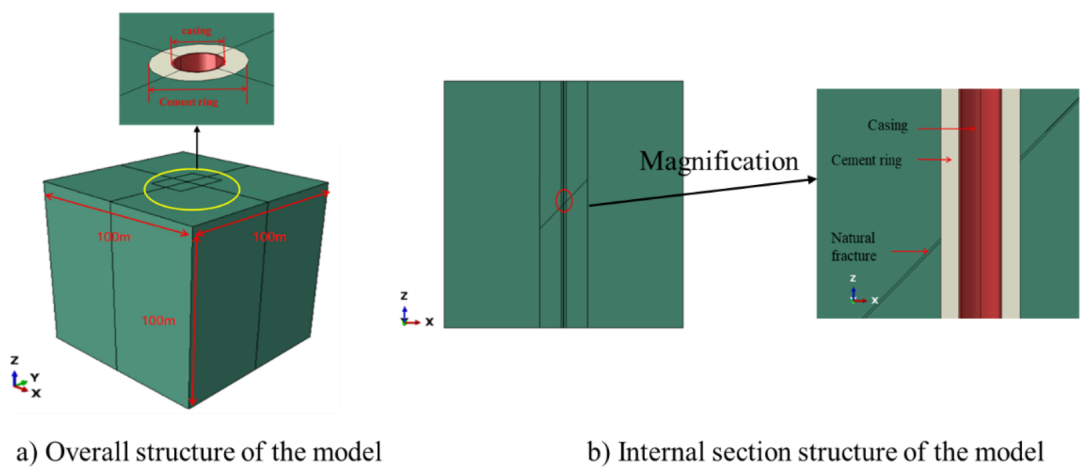


Figure 2—Integrated three-dimensional fracture slip finite element model of formation - cement ring – casing.

Model Verification

During the fracturing of Well W202H-1, bridge plug pumping encountered obstruction. Ant-tracking interpretation revealed the presence of a natural fracture intersecting the wellbore at the obstruction point, with the fracture measuring 150m in length and oriented at N50°E. The wellbore was located approximately 30m from the fracture center. Caliper logging indicated a maximum CD of 31.8m(Lu et al. 2021). Based on the well's geological data (Table 2), a fracture slip model was established, which simulated a maximum CD of 30.58mm (Fig. 3). The relative error of 5.5% demonstrates the model's good reliability and its capability to accurately reflect CD magnitude.

Table 2—Engineering geological basic parameters of Well W202H-1.

Parameter	Value	Parameter	Value
Natural fracture length /m	26.6	Cement ring diameter /mm	215.9
Casing wall thickness/mm	12.7	Young's modulus of rock /GPa	43.5
Young's modulus of cement ring/GPa	20	Cement ring Poisson's ratio	0.25
Young's modulus of the casing/GPa	200	Casing Poisson's ratio	0.3
Maximum horizontal principal stress/MPa	72	Horizontal minimum principal stress/MPa	60
Casing diameter/mm	139.7	Casing density/kg/m³	7800
Rocks Poisson's ratio /dimensionless	0.22	Vertical principal stress/MPa	64
Cement ring density /kg/m³	1800		

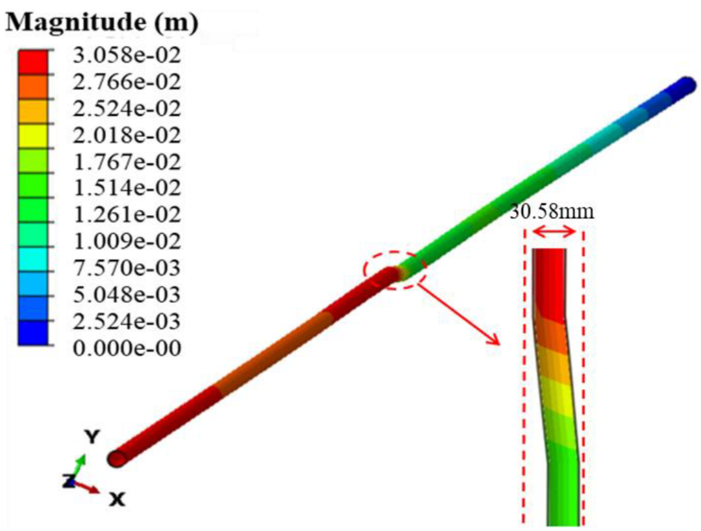


Figure 3—Numerical simulation results of verified model.

Analysis of Influencing Factors on Casing Deformation

During hydraulic fracturing in shale gas wells, CD is closely related to fracturing parameters and associated formation conditions. The fracturing process in shale reservoirs involves various geological, engineering, and environmental factors, which can be broadly categorized into geological factors and engineering factors. Based on the established finite element model of fracture slipinduced CD, this study analyzes the influence of fracture parameters (fracture angle, fracture length), in-situ stress parameters (stress difference and elastic modulus), and operational parameters (fluid pressure within fractures) on CD magnitude to elucidate the lawsofCD.

Inclination Angle of Natural Fractures

The inclination angle of natural fractures is one of the key geological factors influencing CD, as it determines the relative spatial relationship between the wellbore and natural fractures, as well as the direction and magnitude of shear loading on the casing. Therefore, it is essential to analyze the effect of natural fracture dip angle on CD magnitude. Simulations were conducted with natural fracture dip angles set at 30°, 45°, 60°, 75°, and 90°, and the corresponding maximum CD values are presented in Fig. 4.

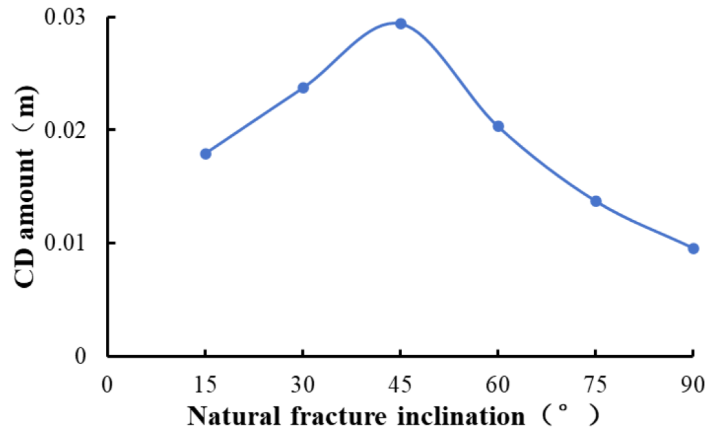


Figure 4—The relationship between the natural fractureinclination and the CD amount.

As can be seen from the figure, when the fracture inclination angle ranges from 30-60°, the CD is more significant. Particularly, when the angle between the maximum principal stress and the normal direction of the shear plane reaches 45°, the risk of fracture slip is highest and the CD reaches its maximum value. This is because, according to the Coulomb failure criterion, when the natural fracture dip angle is 45°, the shear stress reaches its peak value, making the fractures most prone to shear slip. This generates transverse shear forces on the casing, leading to deformation. Therefore, during well trajectory design, fractures with dip angles around 45° should be avoided as much as possible. When unavoidable, drilling parallel to the fracture direction can help reduce the risk of CD during fracturing operations.

Natural Fracture Length

Natural fracture length serves as a critical link between formation mechanical behavior and casing structural response. Increasing fracture length expands the activation range and promotes complex interaction with hydraulic fractures, resulting in asymmetric loading on the casing. Studying the influence of fracture length on CD enables accurate prediction of wellbore integrity risks, optimization of engineering design, and reduction of development costs, ultimately ensuring safe and efficient extraction of oil and gas resources. Simulations were conducted with natural fracture half-lengths ranging from 30 to 80m, and the corresponding maximum CD values are shown in Fig. 5.

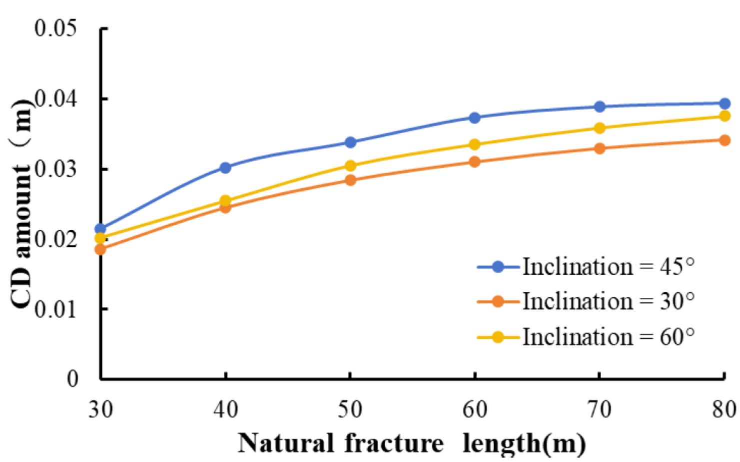


Figure 5—The relationship between the natural fracture length and the CD amount.

As can be observed from the figure, when the fracture angle remains constant, the CD shows a proportional relationship with fracture length—longer fractures result in greater fracture slip displacement and consequently larger CD. This occurs because: (1) increasing fracture length leads to more extensive stress redistribution in surrounding formations, which may directly affect the casing location and create non-uniform loading; (2) longer fractures have larger slip surfaces with reduced frictional resistance, making them more susceptible to shear slip under fluid pressure or in-situ stress disturbances, thereby directly compressing the casing and causing deformation.

In-situ Stress Difference

The horizontal in-situ stress difference serves as the fundamental cause of non-uniform loading on casing. When a difference exists between the maximum and minimum horizontal principal stresses, the casing experiences asymmetric radial compression, leading to deformation. A quantitative study of the influence of horizontal stress difference on CD provides scientific basis for casing design and strength selection, while offering theoretical support for optimizing well trajectories and developing targeted prevention measures. Accordingly, simulations were conducted with horizontal stress differences ranging from 2 to 20 MPa, and the corresponding maximum CD values are shown in Fig. 6.

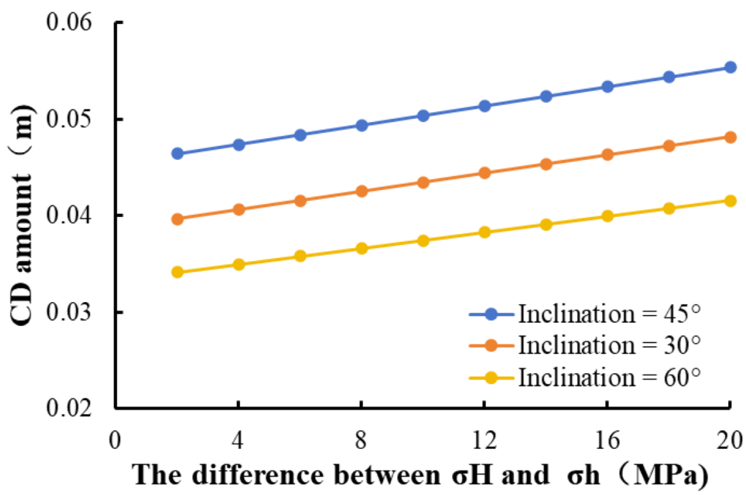


Figure 6—The relationship between the Horizontal ground stress difference and the CD amount.

As shown in the figure, the CD exhibits a positive correlation with the in-situ stress difference, the greater the horizontal stress difference, the more severe the CD. This relationship can be attributed to three main mechanisms: (1) Higher stress differences significantly increase shear stress, making natural fractures more prone to slip and directly shearing the casing; (2) It promote the formation of single, elongated fractures which create asymmetric loading that shears the casing when intersecting the casing at an angle; (3) Elevated stress differences subject the casing to eccentric confining pressure, generating stress concentrations at cement sheath imperfections that induce deformation.

Formation Elastic Modulus

The formation elastic modulus reflects the rock's ability to resist deformation (higher elastic modulus means less rock deformation under external forces; lower elastic modulus makes the rock more prone to deformation). As the structural support of the wellbore, CD is directly influenced by the stress and deformation transfer from surrounding formations. When the formation is subjected to external loads, significant deformation occurs in the rock itself. This deformation is directly transmitted to the casing through the wellbore, causing the casing to bear uneven loads. Therefore, it is necessary to study the influence of formation elastic modulus on CD. Simulations were conducted with formation elastic moduli of 26.6 GPa, 33.6 GPa, 40.6 GPa, 47.6 GPa, 54.6 GPa, and 61.6 GPa, and the corresponding maximum CD values are shown in Fig. 7.

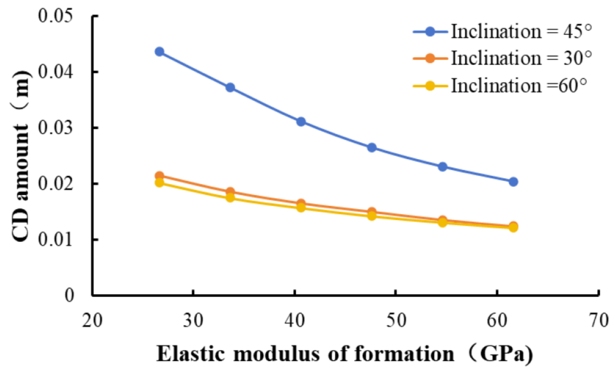


Figure 7—The relationship between the elastic modulus of formation and the CD amount.

As can be observed from the figure, higher elastic modulus of formation corresponds to greater deformation resistance, resulting in smaller fracture slip displacement and reduced CD. This is because rock with higher elastic modulus exhibits stronger deformation resistance, generating smaller strain under identical stress conditions. Consequently, the load transferred from surrounding rock to the casing is significantly reduced, leading to diminished compressive effects on the casing.

Fluid Pressure Within Fracture

Fluid pressure within fracture reflects the intensity of reservoir stimulation during fracturing. Higher injection rates and larger fluid volumes result in increased fluid pressure within fractures. The pressure is transmitted through fracture wall to surrounding formations, altering in-situ stress distribution—elevated fluid pressure compresses rock blocks on both sides of the fracture, creating stress concentration toward the wellbore. The resulting load is then transferred to the casing through the cement sheath, making fluid pressure within fracture a key operational parameter affecting casing stress.To analyze its impact on CD, simulations were conducted with fracture fluid pressures ranging from 30 to 90 MPa, and the corresponding maximum CD values are shown in Fig. 8.

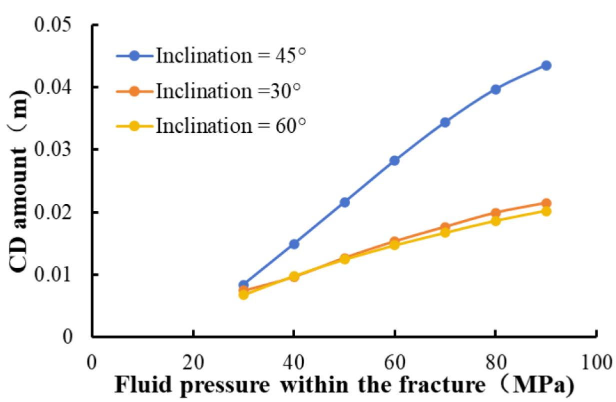


Figure 8—The relationship between the fluid pressure within the fracture and the CD amount.

As shown in the figure, higher fluid pressure within fractures leads to greater fracture slip displacement and consequently increased CD. This phenomenon primarily results from changes in fracture wall stresses. Under initial conditions, the shear stress and frictional forces along fracture walls balance each other, maintaining equilibrium. When internal fluid pressure increases, the supporting effect on fracture walls strengthens, reducing effective stress and consequently decreasing frictional resistance. Once this friction becomes insufficient to counteract the shear stress, fractures are activated and begin to slip. This process causes formation compression toward the wellbore, creating stress concentration that increases radial loading on the casing and ultimately amplifies CD.

CDprevention technology—uncemented fracturing technology with external casing packers

Based on the aforementioned analysis, optimizing well placement (in areas with low in-situ stress difference and high elastic modulus), trajectory design (avoiding fractures with 30-45° inclination angles), and operational parameters (reducing fluid pressure within fracture) can mitigate the shear effects of fracture slip on casing. However, China's shale reservoirs are predominantly located in mountainous regions and generally exhibit deep burial depths, strong heterogeneity, significant horizontal principal stress differences, and complex in-situ stress conditions. Combined with the widespread development of fault and fracture systems, these geological characteristics make it challenging to completely avoid geologically weak zones during well trajectory design, significantly increasing the potential risk of CD.

Furthermore, to enhance shale reservoir development efficiency, the current industry widely adopts "high flow rate + high-intensity" volumetric fracturing techniques, which increase perforation cluster density and stimulation scale to achieve fully stimulation. While this approach boosts production, it inevitably exacerbates formation slip risks, further raising the probability of CD. Although reducing stimulation scale may lower CD risks, it directly compromises well productivity, making it difficult to strike an ideal balance between operational safety and production maximization.

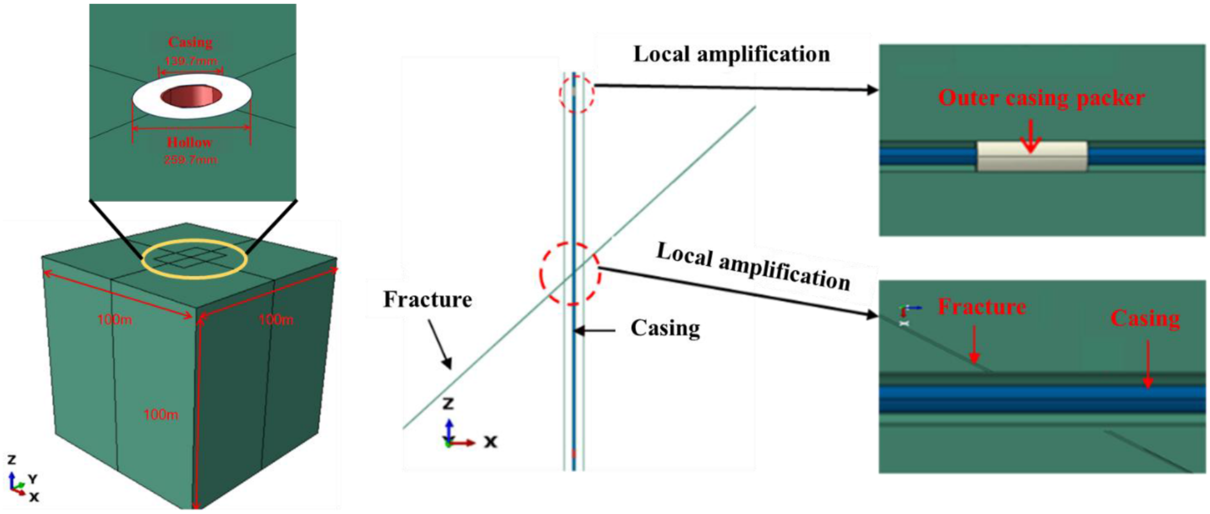
These observations demonstrate that relying solely on passive preventive measures—such as minimizing formation slip forces—proves inadequate for effectively controlling CD while meeting production demands. To address this challenge, this study innovatively proposes a uncemented fracturing technology with external casing packers. The core innovation lies in reserving sufficient slip space between the casing and formation, preventing direct rock-casing contact after fracture slipping and thereby enabling active control of CD. Compared to traditional passive prevention methods, this technology represents a paradigm shift from "passively preventing slip" to "actively managing slip," offering greater initiative and controllability in CD mitigation.

Physical Model Establishment

To validate the effectiveness of this technology, improvements were made based on the integrated formation-cement ring-casing 3D fracture slip finite element model, establishing a finite element model for un-cementing fracturing with external packers to prevent CD. The model dimensions are 100m×100m×100m, with a fracture surface size of 30m×30m×1mm. The casing has an outer diameter of 139.7mm and inner diameter of 118.4mm. Instead of cement ringwith full well sectionbetween the formation and casing, localized packers are used. Two packers are symmetrically installed outside the casing relative to the fracture, with a vertical spacing of 20m from the central fracture (Fig. 9). This design serves two purposes: (1) maintaining a gap between the formation and casing to accommodate fracture slip, (2) providing support for the casing and segmentation. The basic parameters of the model are shown in Table 3.

Table 3—Parameters of the model.

Parameter	Value	Parameter	Value
Young's modulus of formation /GPa	26.6	Rocks Poisson's ratio	0.26
Young's modulus of the packer/GPa	21	Packer Poisson's ratio	0.3
Young's modulus of the casing/GPa	200	Casing Poisson's ratio	0.3
Maximum horizontal principal stress/MPa	85	Horizontal minimum principal stress/MPa	79
fluid pressure within fracture/MPa	88	Natural crack inclination angle/°	45
Rock density/kg/m³	2500	Vertical principal stress /MPa	81
Packer density/kg/m³	7800	Natural crack length /m	80
Casing density/kg/m³	7800		



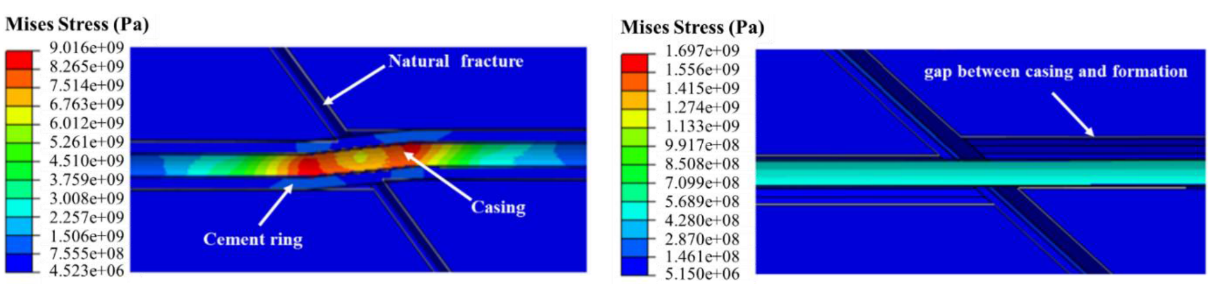
a) Overall structure of the model

b) Internal section structure of the model

Figure 9—Finite element model of non-cementing fracturing technology with outer casing packer.

Feasibility Analysis

Based on the aforementioned model, simulations were conducted to compare casing stress and deformation magnitude under two completion methods—cementing in fullwell section and cemented with external casing packer under conditions of 45° natural fracture inclination angle and 80m fracture length, with results shown in Fig. 10, Fig. 11. Fig. 10 demonstrates the casing load distribution comparison: the maximum load in fullwell section cementing(901.6MPa) was significantly higher than in packer-based cementless completion (106.7MPa), representing an 81.2% reduction that effectively decreases the transfer of formation slip forces to the casing. Analysis of casing cross-sectional deformation near the fracture revealed that full well section cementing resulted in severe deformation of 18.92~45.42mm (15.9~38.4% diameter reduction), while theuncemented fracturingwith external casing packer showed minimal deformation of only 0.23~0.263mm (0.2% diameter reduction). Both the load-bearing capacity and cross-sectional deformation measurements confirm that the uncemented fracturing technology with external casing packers can effectively reduce both the transfer of formation slip forces to the casing and the deformation magnitude.



a) Casing cementing fracturing technology

b) non-cementing fracturing technology with outer casing packer

Figure 10—The load distribution of the casing under different modes.

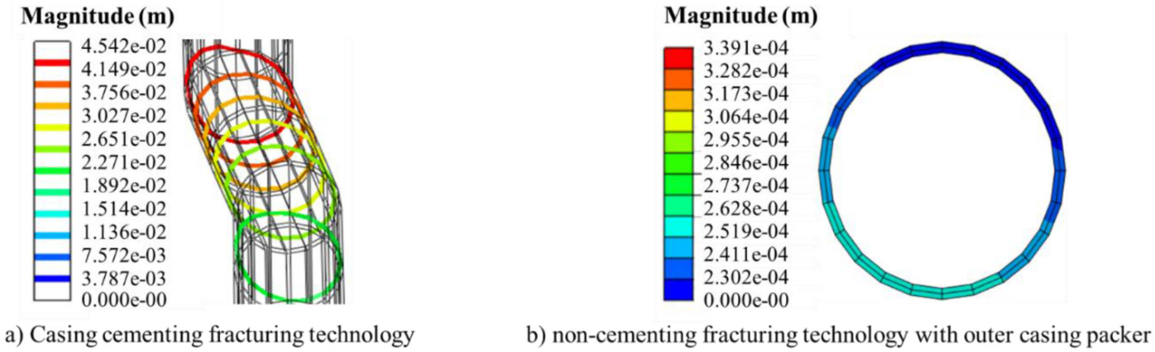


Figure 11—The deformation magnitude of the casing section under different modes.

Field Application

During the perforation operation in Well HX-1 of Block W202, bridge plug encountered obstruction at 3090m depth, with logging measurements showing 31.8mm of CD. Using this well's data, simulations comparing CD behavior between full well section cementing and packer-based cementless fracturing were conducted (Fig. 12). The full well section cementing simulation yielded 30.58mm CD (3.8% error compared to field data) with 33.75mm of fracture slip displacement on both sides. Based on this slip distance, a 34mm gap was designed between casing and formation, while external packers were installed on both sides of the fracture for localized cementing according to the staged design. The simulation of cementless fracturing with external packers showed maximum CD of only 0.22mm, which is negligible for subsequent operations. Field application studies confirm the effectiveness of uncemented fracturing technology with external casing packers in preventing CD.

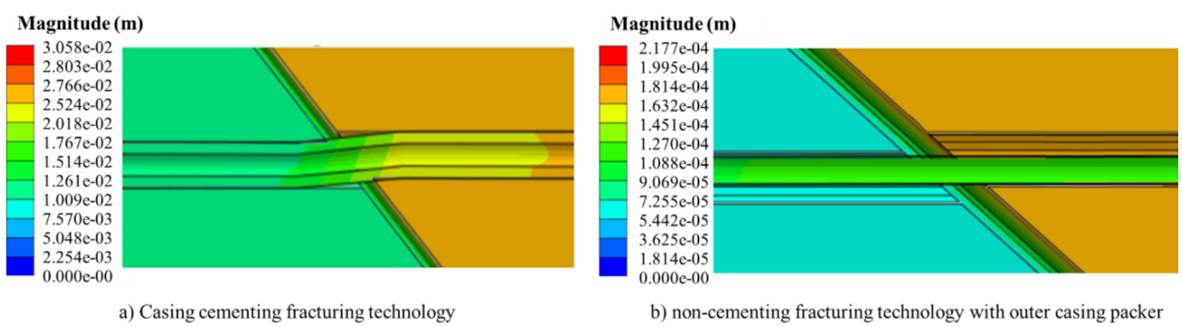


Figure 12—The deformation magnitude of thecasing cementing fracturing technology and non-cementing fracturing technology with outer casing packer.

Conclusion

This study established an integrated formation-cement sheath-casing 3D finite element model to analyze the influence of geological and engineering factors on CD. Based on the CD mechanism and the limitations of existing prevention technologies of CD, we innovatively proposed a uncemented fracturing technology with external casing packers and validated its effectiveness through finite element modeling of packer-based cementless fracturing. The main conclusions are as follows:

1. The primary cause of CD lies in the fracturing process where fluid infiltration into natural fractures and other weak planes activates fracture slip. The resulting shear forces acting on the casing leading to deformation.
2. CD initially increases then decreases with increasing fracture inclination angle, peaking at 45°. Deformation escalates with fracture length, in-situ stress difference, and operational intensity, while decreasing with higher formation Young's modulus.

3. The uncemented fracturing technology with external casing packersenables a paradigm shift from "passively preventing fracture slip" to "actively controlling slip" by maintaining sufficient clearance between casing and formation. This prevents direct rock-casing contact after fracture slipping, achieving proactive deformation control.
4. Compared to full well sectioncementing, the uncemented fracturing technology with external casing packers reduces maximum casing load by 81.2% and diameter reduction by approximately 27%, demonstrating effective prevention of CD.

References

Bois, A., Garnier, A., and Rodot, F. et al 2011. How to prevent loss of zonal isolation through a comprehensive analysis of microannulus formation. *SPE Drill Complet* **26** (01): 13–31.

Guo, X., Li, J., and Liu, G. et al 2018. Research on casing deformation for shale gas wells based on focal mechanism. *Fault-Block Oil & Gas Field* **25** (05): 665–669.

Hu, M., Hongkui, G., and Yuan, Y. et al 2022. A new insight into casing shear failure induced by natural fracture and artificial fracture slip. *Eng Fail Anal* 137. https://doi.org/10.1016/j.engfailanal.2022.106287.

Hu, W. and Bao, J. 2013. To explore the way of chinese-style shale gas development. *Natural Gas Industry* **33** (01): 1–7. https://doi.org/10.3787/j.issn.1000-0976.2013.01.001.

Li, G., Lei, Z., and Dong, W. et al 2022. Progress, challenges and prospects of unconventional oil and gas development of CNPC. *China Petroleum Exploration* **27** (01): 1–11. https://doi.org/10.3969/j.issn.1672-7703.2022.01.001.

Li, J., Xi, Y., and Fu, Y. et al 2017. Improving shale gas well integrity by staging cementing. *Drilling & Production Technology* **40** (04): 21–24. https://doi.org/10.3969/J.ISSN.1006-768X.2017.04.08.

Li, N., Chen, and Jiang et al 2025. Research on the mechanism and influencing factors of deep shale casing deformation. *Special Oil & Gas Reservoirs:* 1–9.

Lin, H., Mao, L., and Mai, Y. 2024. Influence of multistage fracturing in shale gas wells on the casing deformation of horizontal wells. *Pet Sci Technol* **42** (1): 56–80. https://doi.org/10.1080/10916466.2022.2108836.

Liu, H., Liu, W., and Wang, S. et al 2020. Research status and development trend of mechanisms of casing failure caused by volume fracturing in horizontal wells. *Journal of China University of Petroleum(Edition of Natural Science)* **44** (06): 53–62. https://doi.org/10.3969/j.issn.1673-5005.2020.06.007.

Liu, J. 2021. Hydraulic fracturing technology and application of shale gas well. *Chemical Engineering & Equipment* (06): 105–106. https://doi.org/10.19566/j.cnki.cn35-1285/tq.2021.06.050.

Liu, K., Taleghani, A. D., and Gao, D. 2019. Calculation of hydraulic fracture induced stress and corresponding fault slippage in shale formation. *Fuel (Lond)* 254. https://doi.org/10.1016/j.fuel.2019.05.108.

Lu, Q., Liu, Z., and Guo, J. et al 2021. Hydraulic fracturing induced casing shear deformation and a prediction model of casing deformation. *Petroleum Explor Dev* **48** (02): 394–401. https://doi.org/10.11698/PED.2021.02.16.

Meng, H., Ge, H., and Fu, D. et al 2020. Numerical investigation of casing shear deformation due to fracture/fault slip during hydraulic fracturing. *Energy Sci Eng* (No.10): 3588–3601. https://doi.org/10.1002/ese3.766.

Shi, B., Chen, M., and Rao, X. et al 2018. Analysis and understanding of casing damage in shale gas well. *Steel Pipe* **47** (03): 66–71. https://doi.org/CNKI:SUN:GAGU.0.2018-03-019.

Shi, C., Wang, A., and Zhu, X. et al 2023. Casing shear deformation mechanism and prevention measures of multifault slip induced by shale gas volume fracturing. *J Pipeline Syst Eng Pract* **14** (4). https://doi.org/10.1016/j.engfailanal.2022.106287.

Wang, G. and Zhang, C. 2022. Shale gas exploitation technology and suggestions for shale gas development in china. *Petrochemical Industry Technology* **29** (02): 64–65.

Wang, H., Mao, L., and Zhu, J. et al 2025. Adjacent well casing damage caused by fluid pressure intrusion in multistage fracturing. *Pet Sci Technol* **43** (13): 1423–1443.

Xi, Y., Yao, Y., and Chen, J. et al 2023. Study on reducing casing shear deformation by using composite cement slurry column structure. Presented at the 57th U.S. Rock Mechanics/Geomechanics Symposium, Atlanta, Georgia, USA, 2023-6-25. https://doi.org/https://doi.org/10.56952/ARMA-2023-0619.

Xu, Z., Yin, F., and Han, L. et al 2018. Mechanism study of casing deformation in multistage hydraulic fracturing shale reservoir. Presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Houston, Texas, USA, 2018-7-23. https://doi.org/https://doi.org/10.15530/URTEC-2018-2896020.

Yin, F., Han, L., and Yang, S. et al 2018. Casing deformation from fracture slip in hydraulic fracturing. *J Pet Sci Eng* 166. https://doi.org/10.1016/j.petrol.2018.03.010.

Yuan, G., Wang, X., and Lei, Q. et al 2023. Mechanism of casing deformation induced by shale gas well fracturing and its physical simulation analysis. *Natural Gas Industry* **43** (11): 137–145.

Zhang, H., Tong, H., and Zhang, P. et al 2023. How can casing deformation be prevented during hydraulic fracturing of shale gas?—a case study of the weiyuan area in sichuan, china. *J Pet Sci Eng* 221. https://doi.org/10.1016/j.petrol.2022.111251.

Zhao, C., Li, J., and Liu, G. et al 2019. Analysis of the influence of cement sheath failure on sustained casing pressure in shale gas wells. *J Nat Gas Sci Eng* **66**: 244–254.

Zhou, Y. 2022. Research on stimulation technology of shale gas fracturing. *China Petroleum and Chemical Standard and Quality* **42** (04): 167–168.